

Juan Pablo Hidalgo

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Profile

I am PhD student interested in the origin of the large-scale magnetic fields in early-type stars. I explore different scenarios that can lead to the formation and evolution of these magnetic fields using magnetohydrodynamic simulations. I am also interested in the effect of magnetic fields on the habitability of exoplanets.

Education

PhD in Astronomy, Astrophysics and Space Science

- Sapienza Università di Roma, 40th cycle student ([link](#))
Supervisor: Prof. Dominik Schleicher, co-supervisor: Prof. Petri Käpylä

Rome, Italy
Nov 2024 – now

MSc in Astronomy

- Universidad de Concepción (Final grade 6.8/7.0)
Supervisor: Prof. Dominik Schleicher
- Thesis: *Origin of magnetism in early-type stars. Protostellar magnetic fields and core dynamo* ([link](#))

Concepción, Chile
Mar 2022 – Nov 2023

BSc in Astronomy

- Universidad de Concepción (Final grade 6.5/7.0)

Concepción, Chile
Mar 2018 – Dec 2021

Basic and Secondary Education

- Colegio Academia Tarapacá

Iquique, Chile
Mar 2005 – Dec 2016

Scholarships and projects

ANID/DOCTORADO BECAS CHILE 72240057

- Chilean funding for a PhD abroad ([page 15, in Spanish](#))

PhD Scholarship
Nov 2024 – now

ISCRA type B project: Evolution of magnetic fields in pre-main-sequence intermediate-mass stars

- 3 317 760 core hours in G100

HPC project
June 2026 – now

ISCRA type C project: Magnetic fields in pre-main-sequence Herbig Ae stars

- 100 000 core hours in G100

HPC project
Sep 2025 – June 2026

ANID Project QUIMAL 220002: The next step for astrophysical supercomputing in Chile: a powerful GPU-server for fluid and stellar dynamics (PI: Prof. Dominik Schleicher)

- Magnetohydrodynamic simulations were performed with high-power graphics cards

Technical staff
Mar 2024 – Aug 2024

ANID Project FB210003: Centro de Excelencia en Astrofísica y Tecnologías Afines (CATA)

- Numerical simulations of the core dynamo in A-type stars were performed. Dipolar magnetic field configurations were implemented in the PENCIL CODE

Technical staff
May 2023 – Jun 2024

ANID Fondecyt regular 1201280: Eclipsing time variations in post-common-envelope binaries: planetary hypothesis vs magnetic activity (PI: Prof. Dominik Schleicher)

- MHD stellar dynamo simulations

Thesis student
Apr 2023 – Nov 2023

Beca Adelina Gutiérrez: Development of an A-type star model

- Funding from The Chilean Astronomical Society (SOCHIAS) ([link, in Spanish](#))

Additional support
Apr 2023

Publications

- Tidal locking as a negative feedback on Earth-like planetary dynamos: Consequences for magnetic shielding and habitability** Aug 2026
J. P. Hidalgo, & D. R. G. Schleicher
Astronomy & Astrophysics, 712, A117 ([link](#))
- The impact of stellar winds and tidal locking effects on the habitability of Earth-like exoplanets around M-dwarf stars** Dec 2025
J. P. Hidalgo, D. R. G. Schleicher, & D. P. González
Astronomy & Astrophysics, 704, A48 ([link](#))
- Shaping core dynamos in A-type stars: The role of dipolar fossil fields** Jul 2025
J. P. Hidalgo, P. J. Käpylä, D. R. G. Schleicher, C. A. Ortiz-Rodríguez & F. H. Navarrete
Astronomy & Astrophysics, 699, A250 ([link](#))
- Magnetohydrodynamic simulations of A-type stars: Long-term evolution of core dynamo cycles** Nov 2024
J. P. Hidalgo, P. J. Käpylä, D. R. G. Schleicher, C. A. Ortiz-Rodríguez & F. H. Navarrete
Astronomy & Astrophysics, 691, A326 ([link](#))
- Survival of fossil fields during the pre-main sequence evolution** Oct 2023
D. R. G. Schleicher, *J. P. Hidalgo* & D. Galli
Astronomy & Astrophysics, 678, A204 ([link](#))
- Simulations of dynamo action in slowly rotating M dwarfs: Dependence on dimensionless parameters** Oct 2023
C. A. Ortiz-Rodríguez, P. J. Käpylä, F. H. Navarrete, D. R. G. Schleicher, *J. P. Hidalgo* & B. Toro.
Astronomy & Astrophysics, 678, 82 ([link](#))

Conferences

- Cool Stars 23** Tokyo, Japan
Jun 15 – 19, 2026
• TOC Ariake. [Poster](#)
• [Conference proceeding](#)
- Numerical astrophysics meeting** Hamburg, Germany
Sep 10 – 12, 2025
• Hamburg Observatory. [Talk](#)
- The 2024 Pencil Code User Meeting** Barcelona, Spain
Sep 23 – 27, 2024
• Universitat Autònoma de Barcelona. [Talk](#)
- XVII Latin American Regional IAU Meeting** Montevideo, Uruguay
Nov 27 – Dec 1, 2023
• Centro de Conferencias of the Intendencia de Montevideo. [Poster](#)
- 65th Annual Meeting of the Argentine Astronomy Association** San Juan, Argentina
Sep 18 – 22, 2023
• Universidad Nacional de San Juan. [Talk](#)
• [Conference proceeding](#)
- 2nd Annual Meeting of the CATA Astrophysics Center** Santiago, Chile
Aug 01 – 03, 2023
• Hotel Pullman Santiago El Bosque.
- XVIII SOCHIAS meeting** Temuco, Chile
Mar 13 – 16, 2023
• Universidad de La Frontera. [Talk](#)
- Theoretical astrophysics meeting (Area 7)** Concepción, Chile

- Universidad de Concepción. Talk

Nov 07 – 08, 2022

64th Annual Meeting of the Argentine Astronomy Association

Buenos Aires, Argentina
Sep 19 – 23, 2022

- Universidad de Buenos Aires. [Poster](#) 
- [Conference proceeding](#) 

Internships and seminars

Nicolaus Copernicus Astronomical Center (CAMK PAN)

Warsaw, Poland
Mar 11, 2026

- Visit + Colloquium. Slides [here](#) 

Institute for Solar Physics (KIS)

Freiburg, Germany
Jun 10 – 14, 2025

- Supervised by Dr. Petri Käpylä.
- Visit + seminar talk (KIS Colloquium). Slides [here](#) 
- Development of a pre-main-sequence star-in-a-box setup

INAF Osservatorio Astrofisico di Arcetri

Florence, Italy
Dec 12 – 13, 2024

- Supervised by Dr. Daniele Galli
- Visit + seminar talk

Institute for Solar Physics (KIS)

Freiburg, Germany
Jun 19 – Jul 09, 2023

- Supervised by Dr. Petri Käpylä.
- Development of the *zoom setup* presented [here](#) 

Hamburg Observatory

Hamburg, Germany
Jul 10 – 16, 2023

- Supervised by the group of Prof. Robi Banerjee.
- Visit + seminar talk

Teaching and outreach

Outreach talk: *El Gran Apagón: Explorando el fin de la Vida Estelar* (highschool level)

Concepción, Chile
Mar 19, 2024

- Instituto de Humanidades de Concepción

Teaching assistant for the Cosmology course

Concepción, Chile
2023 – 2024

- Course for 1st-year master students at Universidad de Concepción
- Weekly exercise blocks, and homework correction

Teaching assistant for the Stellar Astrophysics course

Concepción, Chile
2022 – 2023

- Course for 3rd-year undergraduate students at Universidad de Concepción
- Weekly exercise blocks, and homework correction

Additional courses

[HYBRID] High-Performance Computing (HPC) Visualisation workshop

Barcelona, Spain
May 14 – 15, 2025

- Barcelona Supercomputing Center (online participation)

Academic Writing and Effective Communication Workshop

Concepción, Chile
Aug 16 – 18, 2022

- Universidad de Concepción (Dirección de Postgrado) ([link](#), [in Spanish](#) 

Intellectual Property Workshop

Concepción, Chile
Oct 27, 2022

- Universidad de Concepción (Dirección de Postgrado)

Astrophysical Dynamos

Online
Jan 24 – 28, 2022

- Online lectures by Dr. Petri Käpylä

Spring School in Applied Mathematics

- Universidad del Bío-Bío ([link](#), [in Spanish](#) )

Concepción, Chile

Oct 18 – 22, 2021

Introduction to the publication of a scientific article

- IOPscience ([link](#), [in Spanish](#) )

Online

Jun 04, 2021

Languages

Spanish (native), **English** (C1, TOEFL iBT), **Polish** (basic), **Italian** (basic).